SOLVED PRACTICE WORKBOOK

General Science: Chemistry Fundamentals - Solved Practice Workbook

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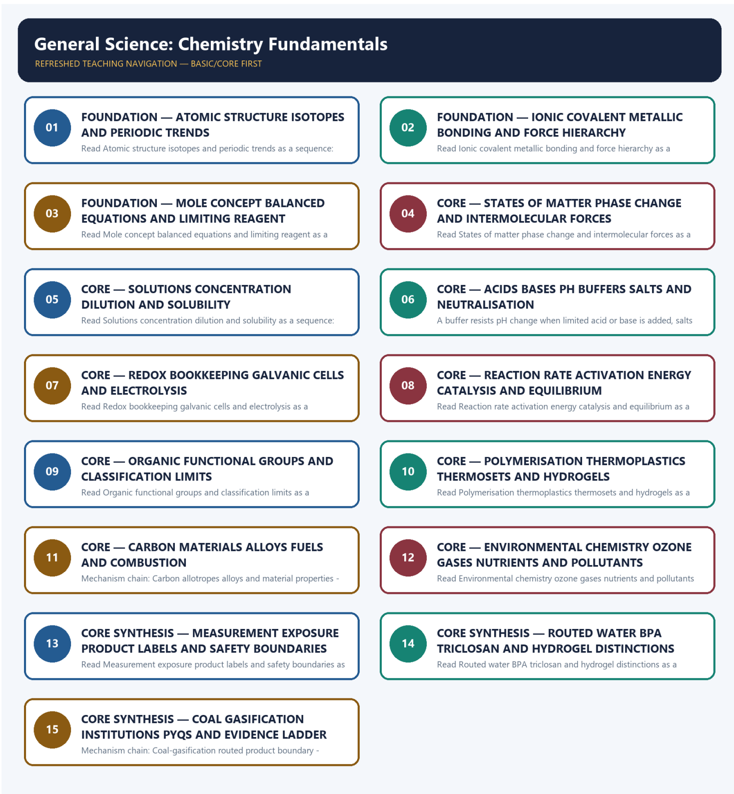

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General Science: Chemistry Fundamentals - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Atomic identity and isotope boundary?

A. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. B. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: A. Explanation: An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Atomic identity and isotope boundary?

A. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: B. Explanation: An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Atomic identity and isotope boundary without changing its institution, unit or status?

A. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. B. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. C. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: C. Explanation: An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Atomic identity and isotope boundary?

A. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. B. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. C. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. D. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number.

Answer: D. Explanation: An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Periodic trends and qualified comparison?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance.

Answer: A. Explanation: The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Periodic trends and qualified comparison?

A. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. B. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. C. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: B. Explanation: The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Periodic trends and qualified comparison without changing its institution, unit or status?

A. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. D. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Answer: C. Explanation: The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Periodic trends and qualified comparison?

A. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. B. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. C. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. D. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence.

Answer: D. Explanation: The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Ionic covalent metallic and intermolecular distinction?

A. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. B. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. C. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. D. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Answer: A. Explanation: Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Ionic covalent metallic and intermolecular distinction?

A. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. B. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. C. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. D. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration.

Answer: B. Explanation: Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Ionic covalent metallic and intermolecular distinction without changing its institution, unit or status?

A. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Answer: C. Explanation: Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Ionic covalent metallic and intermolecular distinction?

A. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. B. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. C. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. D. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance.

Answer: D. Explanation: Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Mole and stoichiometric accounting?

A. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. D. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance.

Answer: A. Explanation: The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Mole and stoichiometric accounting?

A. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. B. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. C. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. D. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration.

Answer: B. Explanation: The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Mole and stoichiometric accounting without changing its institution, unit or status?

A. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. D. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable.

Answer: C. Explanation: The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Mole and stoichiometric accounting?

A. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. B. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. C. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: D. Explanation: The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies States phase change and intermolecular forces?

A. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. D. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Answer: A. Explanation: Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of States phase change and intermolecular forces?

A. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. B. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. C. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. D. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.

Answer: B. Explanation: Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses States phase change and intermolecular forces without changing its institution, unit or status?

A. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. B. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. C. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. D. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable.

Answer: C. Explanation: Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about States phase change and intermolecular forces?

A. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. B. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. C. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. D. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance.

Answer: D. Explanation: Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Solutions concentration and solubility boundary?

A. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. B. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. C. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. D. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable.

Answer: A. Explanation: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Solutions concentration and solubility boundary?

A. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. B. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. C. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. D. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change.

Answer: B. Explanation: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Solutions concentration and solubility boundary without changing its institution, unit or status?

A. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. B. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. C. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. D. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.

Answer: C. Explanation: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Solutions concentration and solubility boundary?

A. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. B. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. C. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. D. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Answer: D. Explanation: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Acids bases neutralisation and pH?

A. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. B. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. C. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. D. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change.

Answer: A. Explanation: An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Acids bases neutralisation and pH?

A. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. B. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. C. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. D. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.

Answer: B. Explanation: An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Acids bases neutralisation and pH without changing its institution, unit or status?

A. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. B. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. C. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. D. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.

Answer: C. Explanation: An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Acids bases neutralisation and pH?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. C. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. D. An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration.

Answer: D. Explanation: An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Buffers salts and neutral-point boundary?

A. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. B. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. C. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. D. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.

Answer: A. Explanation: A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Buffers salts and neutral-point boundary?

A. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. B. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. C. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. D. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.

Answer: B. Explanation: A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Buffers salts and neutral-point boundary without changing its institution, unit or status?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. C. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. D. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Answer: C. Explanation: A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Buffers salts and neutral-point boundary?

A. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. D. A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable.

Answer: D. Explanation: A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Oxidation reduction and electrochemical cells?

A. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. B. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. C. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. D. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change.

Answer: A. Explanation: Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Oxidation reduction and electrochemical cells?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. C. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. D. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.

Answer: B. Explanation: Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Oxidation reduction and electrochemical cells without changing its institution, unit or status?

A. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. D. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Answer: C. Explanation: Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Oxidation reduction and electrochemical cells?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. C. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. D. Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.

Answer: D. Explanation: Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Reaction rate catalyst and equilibrium boundary?

A. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. D. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Answer: A. Explanation: Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Reaction rate catalyst and equilibrium boundary?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. C. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: B. Explanation: Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Reaction rate catalyst and equilibrium boundary without changing its institution, unit or status?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. C. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. D. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone.

Answer: C. Explanation: Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Reaction rate catalyst and equilibrium boundary?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. C. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. D. Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change.

Answer: D. Explanation: Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Organic functional-group classification?

A. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. D. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Answer: A. Explanation: A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Organic functional-group classification?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. C. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: B. Explanation: A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Organic functional-group classification without changing its institution, unit or status?

A. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. B. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. C. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. D. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Answer: C. Explanation: A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Organic functional-group classification?

A. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. B. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. C. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. D. A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.

Answer: D. Explanation: A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Polymer structure and processing classes?

A. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: A. Explanation: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Polymer structure and processing classes?

A. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. B. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. C. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: B. Explanation: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Polymer structure and processing classes without changing its institution, unit or status?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. C. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. D. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation.

Answer: C. Explanation: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Polymer structure and processing classes?

A. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. D. Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Answer: D. Explanation: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Carbon allotropes alloys and material properties?

A. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. B. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. C. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: A. Explanation: Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. The other options change the scientific category, institutional owner,

measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Carbon allotropes alloys and material properties?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. C. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. D. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Answer: B. Explanation: Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Carbon allotropes alloys and material properties without changing its institution, unit or status?

A. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. D. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.

Answer: C. Explanation: Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Carbon allotropes alloys and material properties?

A. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone.

Answer: D. Explanation: Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Fuels combustion and gasification boundary?

A. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. B. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. C. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. D. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation.

Answer: A. Explanation: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Fuels combustion and gasification boundary?

A. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. B. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. C. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. D. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Answer: B. Explanation: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Fuels combustion and gasification boundary without changing its institution, unit or status?

A. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. B. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. C. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. D. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Answer: C. Explanation: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Fuels combustion and gasification boundary?

A. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. B. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.

Answer: D. Explanation: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Environmental chemistry compartment map?

A. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. D. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Answer: A. Explanation: Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Environmental chemistry compartment map?

A. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. B. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.

Answer: B. Explanation: Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Environmental chemistry compartment map without changing its institution, unit or status?

A. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. B. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. C. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. D. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.

Answer: C. Explanation: Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Environmental chemistry compartment map?

A. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway.

Answer: D. Explanation: Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Measurement exposure and category firewall?

A. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. D. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Answer: A. Explanation: pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Measurement exposure and category firewall?

A. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. B. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Answer: B. Explanation: pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Measurement exposure and category firewall without changing its institution, unit or status?

A. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. B. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. C. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. D. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number.

Answer: C. Explanation: pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic,

biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Measurement exposure and category firewall?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Answer: D. Explanation: pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Hydrogel polymer-network route?

A. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. B. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.

Answer: A. Explanation: A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Hydrogel polymer-network route?

A. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. B. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation.

Answer: B. Explanation: A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Hydrogel polymer-network route without changing its institution, unit or status?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. C. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. D. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number.

Answer: C. Explanation: A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Hydrogel polymer-network route?

A. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. D. A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation.

Answer: D. Explanation: A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Water polarity BPA and triclosan boundaries?

A. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Answer: A. Explanation: Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Water polarity BPA and triclosan boundaries?

A. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. B. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence.

Answer: B. Explanation: Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating

status.

Q71. Which statement uses Water polarity BPA and triclosan boundaries without changing its institution, unit or status?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. D. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance.

Answer: C. Explanation: Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Water polarity BPA and triclosan boundaries?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. C. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. D. Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.

Answer: D. Explanation: Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Coal-gasification routed product boundary?

A. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. B. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. C. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. D. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation.

Answer: A. Explanation: Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Coal-gasification routed product boundary?

A. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. B. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence.

Answer: B. Explanation: Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Coal-gasification routed product boundary without changing its institution, unit or status?

A. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. B. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. C. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. D. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance.

Answer: C. Explanation: Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Coal-gasification routed product boundary?

A. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. B. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Answer: D. Explanation: Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Chemistry-to-policy evidence ladder?

A. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. B. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number.

Answer: A. Explanation: Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Chemistry-to-policy evidence ladder?

A. The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. B. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. C. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: B. Explanation: Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Chemistry-to-policy evidence ladder without changing its institution, unit or status?

A. Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. B. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. C. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. D. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.

Answer: C. Explanation: Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Chemistry-to-policy evidence ladder?

A. Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. B. A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. C. The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. D. Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation.

Answer: D. Explanation: Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2024 hydrogel objective demand, the 2025 coal-gasification objective demand and the 2021 water-polarity, bisphenol-A and triclosan objective demands to this owner. Three representative answer-free cards preserve all five routed objective demands. The 2024 GS-III freshwater question remains primarily owned by Environment and Ecology; this topic supplies only the bounded chemistry of membranes, phase change, ions, concentration, disinfection and by-products.

PYQ DEMAND CARD 1 - 2024 Prelims GS-I

Demand: Assess the routed statements on uses of hydrogels.

Status: Verified routed demand covering 2024 Q36; the official Set-A key is available locally but no option, answer letter or objective answer key is reproduced.

Model solution: Polymer structure and processing classes: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Hydrogel polymer-network route:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 1 - 2024 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Polymer structure and processing classes: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

Hydrogel polymer-network route: A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed statements on uses of hydrogels. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed demand covering 2024 Q36; the official Set-A key is available locally but no option, answer letter or objective answer key is reproduced. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Polymer structure and processing classes: Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Hydrogel polymer-network route:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2024 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2025 Prelims GS-I

Demand: Assess the routed statements on what coal-gasification technology can produce.

Status: Verified routed demand covering 2025 Q45; the official Set-A key is available locally but no option, answer letter or objective answer key is reproduced.

Model solution: Fuels combustion and gasification boundary: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Coal-gasification routed product boundary:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2025 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Fuels combustion and gasification boundary: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Coal-gasification routed product boundary:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed statements on what coal-gasification technology can produce. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed demand covering 2025 Q45; the official Set-A key is available locally but no option, answer letter or objective answer key is reproduced. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Fuels combustion and gasification boundary: Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Coal-gasification routed product boundary:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2025 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2021 Prelims GS-I

Demand: Assess the routed distinctions involving water's dipolar solvent character, bisphenol A in material manufacture and triclosan in personal-care products.

Status: Representative routed card covering 2021 Q71, Q74 and Q75; official keys are unavailable locally and no option, answer letter, product list, safety conclusion or regulatory status is inferred.

Model solution: Solutions concentration and solubility boundary: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Organic functional-group classification:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Measurement exposure and category firewall:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Water polarity BPA and triclosan boundaries:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 3 - 2021 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Solutions concentration and solubility boundary: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Organic functional-group classification:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Measurement exposure and category firewall:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Water polarity BPA and triclosan boundaries:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the routed distinctions involving water's dipolar solvent character, bisphenol A in material manufacture and triclosan in personal-care products. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative routed card covering 2021 Q71, Q74 and Q75; official keys are unavailable locally and no option, answer letter, product list, safety conclusion or regulatory status is inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Solutions concentration and solubility boundary: A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Organic functional-group classification:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Measurement exposure and category firewall:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Water polarity BPA and triclosan boundaries:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 3 - 2021 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain how atomic structure and periodic trends guide comparison of elemental properties. Answer in about 150 words.

Model thesis: Claim: Atomic identity and isotope boundary. **Named evidence/example:** An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Periodic trends and qualified comparison. **Named evidence/example:** The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number.
- The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence.

Qualified conclusion: Claim: Atomic identity and isotope boundary. **Named evidence/example:** An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Periodic trends and qualified comparison. **Named evidence/example:** The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain how atomic structure and periodic trends guide comparison of elemental properties.…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Atomic identity and isotope boundary. **Named evidence/example:** An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Periodic trends and qualified comparison. **Named evidence/example:** The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Atomic identity and isotope boundary. **Named evidence/example:** An atom contains protons and neutrons in its nucleus and electrons outside it; atomic number is the proton count that fixes elemental identity, mass number is protons plus neutrons, isotopes share atomic number but differ in mass number, isobars share mass number but differ in atomic number, and isotones share neutron number. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Periodic trends and qualified comparison. **Named evidence/example:** The modern periodic table is ordered by atomic number; across a period atomic size generally decreases while ionisation energy and electronegativity generally increase and metallic character generally decreases, whereas down a group atomic size generally increases. These are trends with contextual exceptions, not licence to rank every pair without electronic-structure evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain how atomic structure and periodic trends guide comparison of elemental properties.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish ionic, covalent and metallic bonding and relate intermolecular forces to states and solutions. Answer in about 150 words.

Model thesis: Claim: Ionic covalent metallic and intermolecular distinction. **Named evidence/example:** Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** States phase change and intermolecular forces. **Named evidence/example:** Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Solutions concentration and solubility boundary. **Named evidence/example:** A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance.
- Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance.
- A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong.

Qualified conclusion: Claim: Ionic covalent metallic and intermolecular distinction. **Named evidence/example:** Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** States phase change and intermolecular forces. **Named evidence/example:** Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Solutions concentration and solubility boundary. **Named evidence/example:** A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on “Distinguish ionic, covalent and metallic bonding and relate intermolecular forces to states…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Ionic covalent metallic and intermolecular distinction. **Named evidence/example:** Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** States phase change and intermolecular forces. **Named evidence/example:** Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Solutions concentration and solubility boundary. **Named evidence/example:** A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit,

source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Ionic covalent metallic and intermolecular distinction. **Named evidence/example:** Ionic bonding arises from electron transfer and attraction between oppositely charged ions, covalent bonding from shared electron pairs, and metallic bonding from positive metal centres with delocalised electrons; hydrogen bonding and van der Waals forces are intermolecular or interparticle attractions and must not be relabelled as the primary bond inside every substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** States phase change and intermolecular forces. **Named evidence/example:** Solid, liquid and gas differ in particle arrangement, mobility and separation; melting, freezing, vaporisation, condensation and sublimation are physical changes of state, while stronger intermolecular attraction generally raises the energy needed for separation. A phase change does not by itself create a new chemical substance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Solutions concentration and solubility boundary. **Named evidence/example:** A solution contains solute dispersed at the molecular or ionic scale in a solvent; concentration describes how much solute is present by a stated basis such as amount, mass, volume or proportion, whereas solubility is the equilibrium limit under specified conditions. Dilute is not synonymous with weak, and concentrated is not synonymous with strong. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish ionic, covalent and metallic bonding and relate intermolecular forces to states…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain mole-based stoichiometry, limiting reagent and yield as a disciplined chemical-accounting framework. Answer in about 250 words.

Model thesis: Claim: Mole and stoichiometric accounting. **Named evidence/example:** The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it.
- pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.

Qualified conclusion: Claim: Mole and stoichiometric accounting. **Named evidence/example:** The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain mole-based stoichiometry, limiting reagent and yield as a disciplined chemical-…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Mole and stoichiometric accounting. **Named evidence/example:** The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant,

purity or yield may be assumed unless the question supplies it. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Mole and stoichiometric accounting. **Named evidence/example:** The mole is the amount-of-substance unit used to connect chemical entities with measurable mass; stoichiometry follows a balanced equation to relate reactants and products, the limiting reagent caps the theoretical product, and percentage yield compares actual with theoretical yield. No numerical constant, purity or yield may be assumed unless the question supplies it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain mole-based stoichiometry, limiting reagent and yield as a disciplined chemical-…", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse acids, bases, pH, buffers, redox cells, reaction rates, catalysis and equilibrium through their correct category boundaries. Answer in about 250 words.

Model thesis: Claim: Acids bases neutralisation and pH. **Named evidence/example:** An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Buffers salts and neutral-point boundary. **Named evidence/example:** A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Oxidation reduction and electrochemical cells. **Named evidence/example:** Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reaction rate catalyst and equilibrium boundary. **Named evidence/example:** Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration.
- A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable.
- Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.
- Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change.

Qualified conclusion: Claim: Acids bases neutralisation and pH. **Named evidence/example:** An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Buffers salts and neutral-point boundary. **Named evidence/example:** A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Oxidation reduction and electrochemical cells. **Named evidence/example:** Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reaction rate catalyst and equilibrium boundary. **Named evidence/example:** Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on “Analyse acids, bases, pH, buffers, redox cells, reaction rates, catalysis and equilibrium…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Acids bases neutralisation and pH. **Named evidence/example:** An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Buffers salts and neutral-point boundary. **Named evidence/example:** A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Oxidation reduction and electrochemical cells. **Named evidence/example:** Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reaction rate catalyst and equilibrium boundary. **Named evidence/example:** Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode.

Analysis: Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Acids bases neutralisation and pH. **Named evidence/example:** An acid can donate protons and a base can accept them, while the aqueous school-level definitions refer to hydrogen and hydroxide ions; pH is logarithmic and tracks hydrogen-ion activity, neutralisation reduces acidic and basic character and often forms salt and water, and acid strength must be kept separate from concentration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Buffers salts and neutral-point boundary. **Named evidence/example:** A buffer resists pH change when limited acid or base is added, salts may produce acidic, basic or neutral solutions depending on their constituent ions, and neutral pH is not universally fixed at the familiar school value because water self-ionisation varies with temperature. A buffer resists change; it does not make pH immutable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Oxidation reduction and electrochemical cells. **Named evidence/example:** Oxidation is electron loss or an increase in oxidation number and reduction is electron gain or a decrease; the paired half-processes can convert chemical energy to electrical energy in a galvanic or voltaic cell, while an electrolytic cell uses electrical energy to drive a non-spontaneous chemical change. Electrode signs depend on cell type, but oxidation remains at the anode and reduction at the cathode. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Reaction rate catalyst and equilibrium boundary. **Named evidence/example:** Reaction rate depends on effective collisions and can change with concentration, temperature, surface area and catalysts; a catalyst provides a lower-activation-energy pathway and speeds forward and reverse reactions without shifting the equilibrium position. Equilibrium is dynamic equality of forward and reverse rates, not cessation of molecular change. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Analyse acids, bases, pH, buffers, redox cells, reaction rates, catalysis and equilibrium…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Discuss how functional groups, polymers, carbon allotropes, alloys, fuels and gasification connect molecular structure with industrial application. Answer in about 300 words.

Model thesis: Claim: Organic functional-group classification. **Named evidence/example:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Polymer structure and processing classes. **Named evidence/example:** Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Carbon allotropes alloys and material properties. **Named evidence/example:** Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Fuels combustion and gasification boundary. **Named evidence/example:** Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hydrogel polymer-network route. **Named evidence/example:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Coal-gasification routed product boundary. **Named evidence/example:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status.
- Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories.

- Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone.
- Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean.
- A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation.
- Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions.

Qualified conclusion: Claim: Organic functional-group classification. **Named evidence/example:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Polymer structure and processing classes. **Named evidence/example:** Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Carbon allotropes alloys and material properties. **Named evidence/example:** Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Fuels combustion and gasification boundary. **Named evidence/example:** Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hydrogel polymer-network route. **Named evidence/example:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Coal-gasification routed product boundary. **Named evidence/example:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **discuss** requires a direct position on "Discuss how functional groups, polymers, carbon allotropes, alloys, fuels and gasification…", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Organic functional-group classification. **Named evidence/example:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Polymer structure and processing classes. **Named evidence/example:** Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Carbon allotropes alloys and material properties. **Named evidence/example:** Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Fuels combustion and gasification boundary. **Named evidence/example:** Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hydrogel polymer-network route. **Named evidence/example:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Coal-gasification routed product boundary. **Named evidence/example:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Organic functional-group classification. **Named evidence/example:** A functional group is the reactive structural feature used to classify organic compounds, including alcohol, aldehyde, ketone, carboxylic-acid, amine and ester families; compounds sharing a functional group can show related reaction patterns, but a label alone does not establish exposure, toxicity, biodegradability or regulatory status. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Polymer structure and processing classes. **Named evidence/example:** Polymers are macromolecules built from repeating units; addition polymerisation joins monomers without eliminating a small molecule, condensation polymerisation forms links while eliminating a small molecule, thermoplastics soften on reheating, and thermosets form networks that do not simply remelt. Biodegradable and compostable remain condition-dependent categories. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Carbon allotropes alloys and material properties. **Named evidence/example:** Diamond, graphite, graphene, fullerenes and carbon nanotubes are carbon allotropes whose different structures produce different properties; an alloy combines a metal with other elements to alter strength, corrosion resistance or workability. Material performance follows structure and processing, not element name alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Fuels combustion and gasification boundary. **Named evidence/example:** Combustion is oxidation that releases energy, whereas gasification converts a carbonaceous feedstock under controlled conditions into a gas mixture that can include carbon monoxide and hydrogen; complete and incomplete combustion, calorific usefulness, pollutants and lifecycle burden are separate evaluative axes. Gasification must not be presented as ordinary burning or as automatically clean. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hydrogel polymer-network route. **Named evidence/example:** A hydrogel is a cross-linked hydrophilic polymer network that absorbs and retains water; routed applications include dressings, contact lenses, controlled release and water-management contexts. It is neither merely a liquid nor a universal water-purification material, and an application category does not prove performance in every formulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Coal-gasification routed product boundary. **Named evidence/example:** Coal gasification uses controlled reaction with oxygen or steam to produce synthesis gas chiefly containing carbon monoxide and hydrogen, with composition and by-products dependent on process and feedstock. A possible product stream is not evidence of plant efficiency, carbon capture, commercial viability or lower lifecycle emissions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Discuss how functional groups, polymers, carbon allotropes, alloys, fuels and gasification…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Critically examine how chemistry fundamentals inform environmental governance, consumer-product assessment, water technologies and evidence-based technology policy. Answer in about 300 words.

Model thesis: Claim: Environmental chemistry compartment map. **Named evidence/example:** Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Water polarity BPA and triclosan boundaries. **Named evidence/example:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chemistry-to-policy evidence ladder. **Named evidence/example:** Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway.
- pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary.
- Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence.
- Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation.

Qualified conclusion: Claim: Environmental chemistry compartment map. **Named evidence/example:** Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Water polarity BPA and triclosan boundaries. **Named evidence/example:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chemistry-to-policy evidence ladder. **Named evidence/example:** Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **critically examine** requires a direct position on "Critically examine how chemistry fundamentals inform environmental governance, consumer-…", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Environmental chemistry compartment map. **Named evidence/example:** Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Measurement exposure and category firewall. **Named evidence/example:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Water polarity BPA and triclosan boundaries. **Named evidence/example:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Chemistry-to-policy evidence ladder. **Named evidence/example:** Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Stratospheric ozone absorbs harmful ultraviolet radiation while ground-level ozone is a secondary pollutant; greenhouse gases, ozone-depleting substances, acids, nutrients and persistent chemicals operate through different atmospheric, water or soil mechanisms. The same molecule can have different significance by location, concentration, exposure and reaction pathway. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** pH, concentration, oxidation number, reaction rate, yield, dose, exposure, persistence and emission are different measurements or categories; labels such as natural, organic, biodegradable, antimicrobial, fuel, solvent or pollutant do not independently prove safety, hazard, performance or legal status. State the measured quantity, basis, conditions and evidence boundary. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Water's molecular polarity supports dissolution of many ionic and polar substances but does not make it a literal universal solvent; bisphenol A is associated with manufacture of some polycarbonate and epoxy materials, while triclosan is a synthetic antimicrobial used in some personal-care products. Presence, exposure, product coverage, safety and regulation require separate dated evidence. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Atomic or molecular principle leads to material property, then process choice, industrial system, application and governance consequence; CSIR-NCL, CSIR-CECRI and the metallurgical research ecosystem illustrate distinct chemistry, electrochemistry and materials roles. A research mandate, mission page, negotiating document or named application does not prove deployment, outcome, hazard or regulation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

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Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Critically examine how chemistry fundamentals inform environmental governance, consumer-…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.